

Appendix — full results, robustness and code (companion to the 3-page report)

Out-of-sample on the 2016–2024 test set; ratios are MSE relative to HAR (<1 beats HAR). Machine-readable tables in `outputs/`; full critique in `CRITIQUE.md`.

A1. Full 22-model out-of-sample MSE ratios vs HAR — M_ALL, h=1 (NN columns are the 10-net ensembles)

Stock	LogHAR	LevHAR	SHAR	HARQ	HAR-X	RR	LA	EN	P-LA	A-LA	BG	RF	GB	NN1	NN2	NN3	NN4
AAPL	.944	1.026	1.004	.985	1.015	.987	.941	.953	.929	1.991	1.167	1.069	1.002	.920	.891	.938	.921
AMZN	1.000	1.065	.991	.992	.951	.951	.881	.970	.959	1.525	1.140	.963	.978	.869	.879	.861	.941
JPM	.924	1.049	1.177	1.163	1.119	1.093	.951	.953	.979	2.169	1.054	.952	.955	.977	.912	.950	.944

A1 (cont.) — M_ALL, h=22 (note the fixed-window tree collapse — A3)

Stock	LogHAR	LevHAR	SHAR	HARQ	HAR-X	RR	LA	EN	P-LA	A-LA	BG	RF	GB	NN1	NN2	NN3	NN4
AAPL	.666	.842	1.008	1.025	1.123	1.079	1.021	1.031	.921	1.638	7.314	3.038	2.677	1.067	1.009	.926	.814
AMZN	1.214	.949	.996	1.161	1.141	1.114	.901	1.085	1.115	1.410	6.719	3.885	4.069	1.266	1.129	1.245	1.364
JPM	.630	1.112	.988	1.476	1.494	1.481	.976	1.012	.721	1.694	13.835	5.274	1.625	.901	.860	.716	1.309

A2. Apples-to-apples M_HAR (3 RV lags only), h=1 — only NN ensembles/LogHAR reach HAR; trees lose 4–58%; RR/LA/EN/P-LA tie

Stock	LogHAR	HARQ	RR	LA	EN	P-LA	BG	RF	GB	NN2	NN3	NN4
AAPL	.944	.986	.992	1.023	1.021	1.003	1.578	1.406	1.036	.997	.978	.976
AMZN	1.003	.997	.991	.993	.995	.995	1.420	1.260	1.126	1.046	1.064	1.042
JPM	.916	1.161	.984	.995	.994	.977	1.363	1.231	1.088	.999	.995	.985

A3. Critique evidence (verified; full development in CRITIQUE.md)

- **Path decomposition, h=1 (HAR→best NN).** regularisation +4.7/3.0/4.7pp (AAPL/AMZN/JPM); tree marginal −11.6/+0.7/+0.1pp; deep-NN +17.8/+10.2/+4.0pp; total ML gain 10.9/13.9/8.8pp.
- **Fixed- vs rolling-window trees (AAPL h=22).** RF MSE/HAR = 3.04 fixed → 1.55 rolling; the daily re-fit, not the tree class, drives CSV's long-horizon result.
- **Significance.** MCS (90%) retains 20–21/22 models per stock at h=1, HAR never eliminated; DM finds 0–3 models/stock beating HAR at 5%; under Holm/BH essentially none survive at h=1/h=5.
- **Loss-conditional (QLIKE, h=1, mean over stocks).** LogHAR 0.96, RF 0.99, but LA 1.28, NN3 1.06, NN2 1.17, EN 2.32 — the MSE edge largely reverses under the noise-robust loss.
- **Measurement error (realised kernel vs 5-min RV, JPM h=1).** RK/RV MSE = 0.729 (−27%) — larger than the entire ML-vs-HAR gap.
- **Error correlation.** Every model correlates with HAR's errors at $r \geq 0.79$ (winners ≈ 0.88 –0.99); each winner is HAR + a small orthogonal perturbation ($< \sim 20\%$ of error variance).
- **Regime conditionality (COVID split 2016–2019/2020–2024).** Best-ML edge collapses from ≈ 9 –14% to ≈ 0 –2% (AAPL P-LA −2.1%, AMZN EN −1.7%, JPM none beats HAR).

A4. Variable importance (ALE), AAPL h=1 — reproduces CSV Fig 7: RVD/RVW dominate; IV/VIX weighted higher by RF/NN; EA≈0

feature	HAR-X	EN	RF	NN2
RVD	.316	.382	.204	.040
RVW	.250	.357	.183	.186
IV	.106	.191	.214	.232
VIX	.106	.000	.230	.146
EA	.000	.000	.000	.000

A5. Selected code excerpts (methodological fidelity)

Horizon target (one-step alignment $RV_{\{t-1\}} \rightarrow RV_t$):

```
def make_horizon_target(rv, h):
    return pd.concat([rv.shift(-k) for k in range(h)], axis=1).mean(axis=1).rename('y')
```

HARQ insanity filter (clip to in-sample range, BPQ 2016); positivity filter (CSV p.1691):

```
return np.clip(raw, self.train_min_, self.train_max_)
floor=float(train['y'].min()); pred = pred.clip(lower=floor) # every model
```

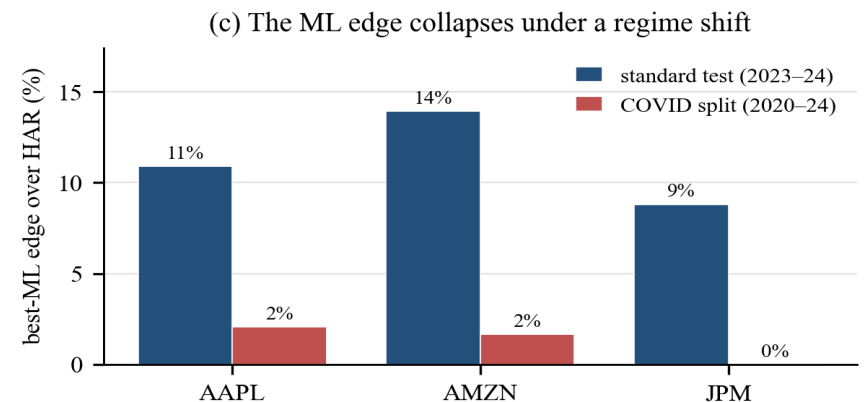
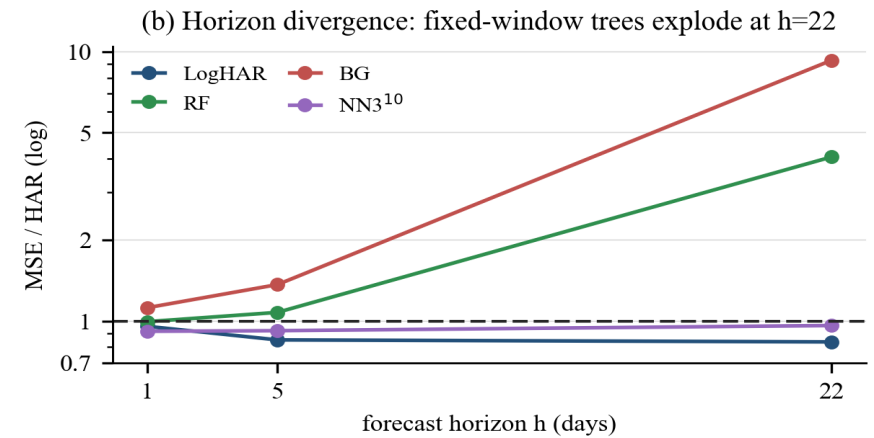
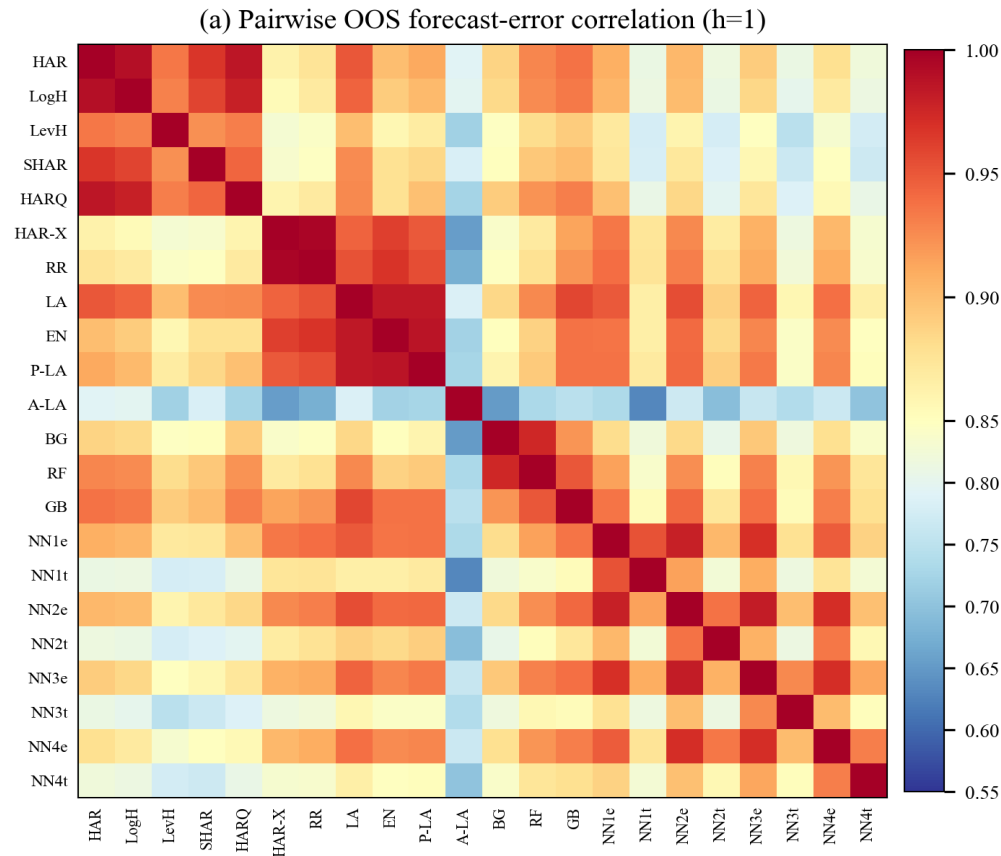
MCS T_max with moving-block bootstrap (Hansen–Lunde–Nason 2011):

```
d_i=Lbar-Lbar.mean(); t=d_i/sd; boot_t=(boot_d-d_i)/sd
pval=np.mean(boot_t.max(axis=1) >= t.max()) # eliminate argmax if pval<=alpha
```

A6. Data & disclosed deviations

5-min RV within-day (no overnight), BNHLS outlier filter, RV/RV±RQ; 70/10/20 chronological split. Deviations: sklearn NN (ReLU, no dropout; early stopping in dropout's place); regularised+trees fixed-window (HAR & NN match CSV); IV proxied by VIX; 100-point λ grid. 37 unit tests pass; python scripts/00_run_full_pipeline.py.

A7. Critique evidence board



(a) Out-of-sample error correlation across the 22 models — every model correlates with HAR at $r \geq 0.79$; only A-LA and the single-seed NNs diverge, so the model space is far narrower than the count suggests. (b) Horizon divergence — fixed-window trees (BG, RF) explode to 5–9× HAR at $h=22$ (log axis) while LogHAR and NN3 stay below HAR; the paper avoids this by rolling BG/RF daily. (c) The best-ML edge over HAR collapses from 9–14% on the standard 2023–24 test to 0–2% on the 2016–2019 / 2020–2024 regime-spanning split.